

Prediction of Chaotic Behavior

Toshiki Oguchi, *Member, IEEE*, and Henk Nijmeijer, *Fellow, IEEE*

Abstract—This paper considers the prediction of chaotic behavior using a master–slave synchronization scheme. Based on the stability theory for retarded systems using a Lyapunov–Krasovskii functional, we derive a sufficient condition for perfect state prediction of the master system via a time-delayed output signal of the slave system. The obtained result is based on the delay-dependent stability of time-delay systems. In addition, we derive an upper bound of the admissible time delay by using linear matrix inequality techniques. Finally, we show the effectiveness of the proposed predictor by two numerical examples.

Index Terms—Complex systems, state prediction, synchronization, time delay.

I. INTRODUCTION

RECENTLY, a great deal of research [1] has been devoted to chaotic systems and complex systems. Especially the synchronization [2]–[4] of coupled chaotic systems has attracted a lot of attention from theoretical and practical viewpoints. Also, as an extension of the synchronization of coupled systems, anticipating synchronization was investigated by Voss [5]. In that paper, the author has shown the possibility of anticipated synchronization of chaotic systems with time delay in a unidirectional coupling configuration and, in addition, numerically illustrated the anticipated synchronization of systems without time delay. The anticipated synchronization can be recognized as a state prediction method which does not require numerical integration. These phenomena are also observed for physical systems. Voss [6] showed an experimental realization of such a phenomenon in an electronic circuit. Masoller [7] has considered the anticipating synchronization of chaotic external-cavity semiconductor lasers with numerical simulation. Sivaprakasam *et al.* [8] have observed the occurrence of anticipating synchronization using chaotic semiconductor diode lasers. These chaotic systems are described by one dimensional systems with time delay in the state, but the anticipated synchronization can occur in more general systems.

In this paper, we consider this behavior as a prediction action of the master system and derive a sufficient condition for perfect state prediction of the master system. The obtained result is based on a delay-dependent stability of time-delay systems. In addition, we derive an upper bound of the available time delay by using the linear matrix inequality (LMI) technique [13], [15].

Manuscript received October 7, 2004; revised February 10, 2005. This work was supported in part by The Ministry of Education, Culture, Sports, Science and Technology, Japan. This paper was recommended by Associate Editor C. T. Lin.

T. Oguchi is with the Department of Mechanical Engineering, Graduate School of Engineering, Tokyo Metropolitan University, Tokyo, 192-0397, Japan (e-mail: oguchi@ctrl.prec.metro-u.ac.jp).

H. Nijmeijer is with the Department of Mechanical Engineering, Eindhoven University of Technology, 5600 MB Eindhoven, The Netherlands.

Digital Object Identifier 10.1109/TCSI.2005.853396

The present paper is organized as follows. In Section II, we propose a new prediction scheme of complex systems which is based upon synchronization. The proposed predictor is designed by using a delay-dependent stability or stabilizability conditions based on a Lyapunov–Krasovskii functional. In Section III, we estimate the available length of the prediction by the proposed scheme. Section IV shows the effectiveness of the proposed predictor by numerical examples.

II. PREDICTION OF CHAOTIC BEHAVIOR

We first review the prediction scheme from [5]. In [5], the author considers the following coupled Ikeda equation:

$$\begin{cases} \dot{x}(t) = -\alpha x(t) - \beta \sin x(t - \tau) \\ \dot{z}(t) = -\alpha z(t) - \beta \sin x(t) \end{cases} \quad (1)$$

where $x, z \in \mathbb{R}$ and $\tau > 0$ denotes a time delay. The dynamics of the prediction error $e(t) \triangleq z(t - \tau) - x(t)$ is given by $\dot{e}(t) = -\alpha e(t)$, and a necessary and sufficient condition for the error e to converge to 0 is that $\alpha > 0$. This idea can be generalized as follows. Consider the following difference-differential equation system:

$$\begin{cases} \dot{\mathbf{x}}(t) = A_0 \mathbf{x}(t) + F(y(t - \tau)) \\ \dot{\mathbf{y}}(t) = h(\mathbf{x}(t)) \end{cases} \quad (2)$$

where $\mathbf{x} \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, $A_0 \in \mathbb{R}^{n \times n}$, $F : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

For the system (2), we consider the following prediction scheme:

$$\dot{\mathbf{z}}(t) = A_0 \mathbf{z}(t) + F(y(t)) \quad (3)$$

in which $\mathbf{z} \in \mathbb{R}^n$. Since the dynamics of the prediction error $\mathbf{e}(t) = \mathbf{z}(t - \tau) - \mathbf{x}(t)$ is described by $\dot{\mathbf{e}}(t) = A_0 \mathbf{e}(t)$, the error converges to zero if and only if matrix A_0 is Hurwitz, i.e., A_0 has all eigenvalues in the open left halfplane. Therefore, if A_0 is a Hurwitz matrix, the output $\mathbf{z}(t)$ of the predictor (3) estimates the future value $\mathbf{x}(t + \tau)$ of the state $\mathbf{x}(t)$.

This prediction method is simple and the convergency of the error dynamics is globally guaranteed, while the master system is required to satisfy the conditions 1) A_0 is a Hurwitz matrix, and 2) the time-delayed terms in the dynamics (2) only depend on the output. These restrictions can be relaxed by adding a coupling term into the slave system, as will be done in the sequel.

We consider the following possibly chaotic or complex system:

$$\Sigma_M : \begin{cases} \dot{\mathbf{x}}(t) = A_0 \mathbf{x}(t) + \sum_{i=1}^m A_i \mathbf{x}(t - \tau_i) + f(\mathbf{x}(t)) \\ y(t) = C \mathbf{x}(t) \\ \mathbf{x}(t) = \varphi(t), \quad t \in [-\max(\tau_i), 0] \end{cases} \quad (4)$$

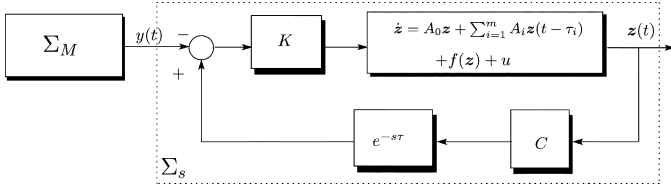


Fig. 1. Configuration of the prediction scheme.

where $\mathbf{x} \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, A and A_i are $n \times n$ matrices, C is a $m \times n$ matrix, τ_i for $i = 1, \dots, m$ are time delays, and $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ denotes a nonlinear term which is continuously differentiable.

For the system (4), we construct the predictor

$$\Sigma_s: \begin{cases} \dot{\mathbf{z}}(t) = A_0 \mathbf{z}(t) + \sum_{i=1}^m A_i \mathbf{z}(t - \tau_i) + f(\mathbf{z}(t)) \\ \quad + K \{ C \mathbf{z}(t - \tau_m) - y(t) \} \\ \mathbf{z}(t) = \mathbf{z}_0, \quad t \in [-2 \max(\tau_i), 0] \end{cases} \quad (5)$$

where $\mathbf{z} \in \mathbb{R}^n$ and $K \in \mathbb{R}^{n \times m}$ is a constant matrix. For the sake of simplicity of notations, we will often suppress the notation of the explicit dependence of time t when no confusion can arise. The configuration of the prediction scheme is shown in Fig. 1.

Then the dynamics of prediction error $\mathbf{e} \triangleq \mathbf{z}_{\tau_m}(t) - \mathbf{x}(t)$, where $\mathbf{z}_{\tau_m}(t) \triangleq \mathbf{z}(t - \tau_m)$, is given by

$$\begin{aligned} \dot{\mathbf{e}} &= A_0 \mathbf{e} + \sum_{i=1}^m A_i \mathbf{e}_{\tau_i} + K C \mathbf{e}_{\tau_m} + f(\mathbf{e} + \mathbf{x}) - f(\mathbf{x}) \\ &\triangleq A_0 \mathbf{e} + \sum_{i=1}^m A_i \mathbf{e}_{\tau_i} + \phi(\mathbf{x}, \mathbf{e}) \end{aligned} \quad (6)$$

with $A_m \triangleq A_m + K C$ and $\phi(\mathbf{x}, \mathbf{e}) = f(\mathbf{e} + \mathbf{x}) - f(\mathbf{x})$. Clearly, $\mathbf{e} \equiv 0$ is a solution of (6), and the system (5) acts as a predictor for (4) if the error dynamics (6) has $\mathbf{e} = 0$ as asymptotically stable equilibrium. In this section, we derive a sufficient condition for the zero solution of (6) to be asymptotically stable.

A. General Approach

We define the Jacobian of $\phi(\mathbf{x}, \mathbf{e})$ with respect to $\mathbf{e}$ at $\mathbf{e} = 0$ by

$$D(\mathbf{x}) \triangleq \left(\frac{\partial \phi(\mathbf{x}, \mathbf{e})}{\partial \mathbf{e}} \right) \Big|_{\mathbf{e}=0} \quad (7)$$

whose components are functions of $\mathbf{x}$. Then the linearization of (6) about $\mathbf{e} = 0$ is given by

$$\dot{\mathbf{e}} = A_0 \mathbf{e} + \sum_{i=1}^m A_i \mathbf{e}_{\tau_i} + D(\mathbf{x}) \mathbf{e}. \quad (8)$$

It is well known that if $\mathbf{e} = 0$ of (8) is asymptotically stable, then $\mathbf{e} = 0$ of (6) is also asymptotically stable [14]. Therefore, using this linearized system, we obtain the following lemma concerning the stability of the trivial solution $\mathbf{e} = 0$ of (6).

Lemma 2.1: Assume the nonempty set $\Omega \subset \mathbb{R}^n$ is such that all trajectories of the system Σ_M converge to it. If there exists a symmetric and positive-definite matrix $P > 0$ and positive-

definite matrices Γ_k , $0 \leq k \leq m$, and Υ of dimension $n \times n$ satisfying

$$\begin{aligned} &(\bar{A} + D(\mathbf{x}))^T P + P(\bar{A} + D(\mathbf{x})) \\ &+ \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T \Gamma_j A_j + D(\mathbf{x})^T \Upsilon D(\mathbf{x}) \right. \\ &\quad \left. + P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) A_i^T P \right) < 0 \end{aligned} \quad (9)$$

for all $\mathbf{x} \in \Omega$, where $\bar{A} = \sum_{i=0}^m A_i$, then the zero solution of (6) is asymptotically stable.

Proof: From Lyapunov's Indirect method for retarded nonlinear systems [9], if the zero solution of the linearized system (8) is asymptotically stable, the zero solution of the original system (6) is also locally asymptotically stable. Therefore, we show that if the inequality (9) holds, then the zero solution of (8) is asymptotically stable. From (8), we have

$$\begin{aligned} \mathbf{e}(t - \tau_i) &= \mathbf{e}(t) - \int_{t-\tau_i}^t \dot{\mathbf{e}}(s) ds \\ &= \mathbf{e}(t) - \int_{t-\tau_i}^t \left\{ A_0 \mathbf{e}(s) + \sum_{i=1}^m A_i \mathbf{e}(s - \tau_i) \right. \\ &\quad \left. + D(\mathbf{x}) \mathbf{e}(s) \right\} ds. \end{aligned} \quad (10)$$

Substituting (10) into (8), we obtain

$$\begin{aligned} \dot{\mathbf{e}}(t) &= A_0 \mathbf{e}(t) + D(\mathbf{x}) \mathbf{e}(t) \\ &+ \sum_{i=1}^m A_i \left\{ \mathbf{e}(t) - \int_{t-\tau_i}^t \left\{ A_0 \mathbf{e}(s) \right. \right. \\ &\quad \left. \left. + \sum_{j=1}^m A_j \mathbf{e}(s - \tau_j) + D(\mathbf{x}) \mathbf{e}(s) \right\} ds \right\} \\ &= \left(A_0 + \sum_{i=1}^m A_i \right) \mathbf{e}(t) + D(\mathbf{x}) \mathbf{e}(t) \\ &- \sum_{i=1}^m \sum_{j=0}^m A_i A_j \int_{t-\tau_i}^t \mathbf{e}(s - \tau_j) ds \\ &- \sum_{i=1}^m A_i \int_{t-\tau_i}^t D(\mathbf{x}) \mathbf{e}(s) ds. \end{aligned} \quad (11)$$

Now we introduce a Lyapunov–Krasovskii functional $V(\mathbf{e}_t)$ as

$$\begin{aligned} V(\mathbf{e}_t) &= \mathbf{e}(t)^T P \mathbf{e}(t) \\ &+ \sum_{i=1}^m \sum_{j=0}^m \int_{\tau_j}^{\tau_i + \tau_j} \int_{t-s}^t \mathbf{e}(u)^T Q_j \mathbf{e}(u) du ds \\ &+ \sum_{i=1}^m \int_0^{\tau_i} \int_{t-s}^t \mathbf{e}(u)^T D(\mathbf{x})^T R D(\mathbf{x}) \mathbf{e}(u) du ds \end{aligned}$$

with $Q_j = A_j^T \Gamma_j A_j$ for $j = 0, \dots, m$ where Γ_j for $j = 0, \dots, m$ and R are symmetric and positive-definite matrices.

Differentiating $V(e_t)$ along the solution of (11), we obtain where the first equation at the bottom of the page. On the other hand, the relations

$$\begin{aligned} & -2 \int_{t-\tau_i-\tau_j}^{t-\tau_j} e(t)^T P A_i A_j e(s) ds \\ & \leq \tau_i e(t)^T P A_i \Gamma_j^{-1} A_i^T P e(t) \\ & \quad + \int_{t-\tau_i-\tau_j}^{t-\tau_j} e(s)^T A_j^T \Gamma_j A_j e(s) ds \\ & -2 \int_{t-\tau_i}^t e(t)^T P A_i D(\mathbf{x}) e(s) ds \\ & \leq \tau_i e(t)^T P A_i \Upsilon^{-1} A_i^T P e(t) \\ & \quad + \int_{t-\tau_i}^t e(s)^T D(\mathbf{x})^T \Upsilon D(\mathbf{x}) e(s) ds \end{aligned}$$

hold for any positive-definite Γ_j and Υ . Substituting the above relations with $Q_j = A_j^T \Gamma_j A_j$ and $R = \Upsilon$, we obtain the inequality, shown in the second equation at the bottom of the page. As a result, if inequality (9) holds, then $\dot{V}(e_t) < 0$ and this means that the zero solution of system (8) is asymptotically stable, and consequently the zero solution of system (6) is also asymptotically stable. $\square$

Using Lemma 2.1, we obtain the following theorem concerning how to choose gain K stabilizing system (6).

Theorem 2.2: Let $\tau_i > 0$ for $i = 1, \dots, m$. If there exist a symmetric and positive-definite matrix $P > 0$ and a matrix Y such that the following holds for all $\mathbf{x} \in \Omega$:

$$\begin{bmatrix} M_{11} + M_{11}^T & M_{12} & M_{13} & M_{14} \\ M_{12}^T & M_{22} & 0 & 0 \\ M_{13}^T & 0 & M_{33} & 0 \\ M_{14}^T & 0 & 0 & M_{44} \end{bmatrix} < 0 \quad (12)$$

$$\begin{aligned} M_{11} &= P(\bar{A} + D(\mathbf{x})) + YC \\ M_{12} &= (A_0^T P, \dots, (A_m^T P + C^T Y^T)) \\ M_{13} &= (P A_1, \dots, (P A_m + YC)) \\ M_{14} &= D(\mathbf{x})^T P \\ M_{22} &= \frac{-1}{\sum_{i=1}^m \tau_i} \text{diag}(P, \dots, P) \\ M_{33} &= \frac{-1}{m+2} \text{diag}\left(\frac{1}{\tau_1} P, \dots, \frac{1}{\tau_m} P\right) \\ M_{44} &= \frac{-1}{\sum_{i=1}^m \tau_i} P \end{aligned}$$

then the zero solution of system (6) is asymptotically stable and $K = P^{-1}Y$ is a stabilizing gain matrix.

Proof: Replacing A_m with $A_m + KC$ in (9), setting $\Gamma_i = \Upsilon = P$ and $Y = PK$, and applying a Schur complement [10], we obtain LMI (12). This completes the proof. $\square$

Remark 2.3: In case that $\tau_i = 0$, by substituting $Y = PK$ into inequality (12), we obtain the following inequality:

$$(\bar{A} + D(\mathbf{x}))^T P + C^T K^T P + P(\bar{A} + D(\mathbf{x})) + PKC < 0. \quad (13)$$

This inequality equals to the (1,1)-element of LMI (12). This means that a feasibility condition of this criterion is that

$$\dot{\mathbf{w}}(t) = (\bar{A} + D(\mathbf{x}))^T \mathbf{w}(t) + C^T u(t)$$

is stabilizable by a static feedback $u = K\mathbf{w}(t)$.

$$\begin{aligned} \frac{d}{dt} V(e_t) &= \dot{e}^T P e + e^T P \dot{e} + \sum_{i=1}^m \sum_{j=0}^m \int_{\tau_j}^{\tau_i+\tau_j} (e(t)^T Q_j e(t) - e(t-s)^T Q_j e(t-s)) ds \\ & \quad + \sum_{i=1}^m \int_0^{\tau_i} (e(t)^T D(\mathbf{x})^T R D(\mathbf{x}) e(t) - e(t-s)^T D(\mathbf{x})^T R D(\mathbf{x}) e(t-s)) ds \\ &= e(t)^T \left\{ \left(\sum_{i=0}^m A_i + D(\mathbf{x}) \right)^T P + P \left(\sum_{i=0}^m A_i + D(\mathbf{x}) \right) \right\} e(t) \\ & \quad - 2 \sum_{i=1}^m \sum_{j=0}^m \int_{t-\tau_i-\tau_j}^{t-\tau_j} e(t)^T P A_i A_j e(s) ds - 2 \sum_{i=1}^m \int_{t-\tau_i}^t e(t)^T P A_i D(\mathbf{x}) e(s) ds \\ & \quad + \sum_{i=1}^m \sum_{j=0}^m \tau_i e^T(t) Q_j e(t) - \sum_{i=1}^m \sum_{j=0}^m \int_{\tau_j}^{\tau_i+\tau_j} e(t-s)^T Q_j e(t-s) ds \\ & \quad + \sum_{i=1}^m \tau_i e(t)^T D(\mathbf{x})^T R D(\mathbf{x}) e(t) - \sum_{i=1}^m \int_0^{\tau_i} e(t-s)^T D(\mathbf{x})^T R D(\mathbf{x}) e(t-s) ds. \end{aligned}$$

$$\begin{aligned} \frac{d}{dt} V(e_t) &\leq e(t)^T \left\{ \left(\sum_{j=0}^m A_j + D(\mathbf{x}) \right)^T P + P \left(\sum_{j=0}^m A_j + D(\mathbf{x}) \right) \right\} e(t) \\ & \quad + e^T(t) \sum_{i=1}^m \tau_i \left\{ \sum_{j=0}^m (A_j^T \Gamma_j A_j + P A_i (\Gamma_j^{-1}) A_i^T P) + P A_i \Upsilon^{-1} A_i^T P + D(\mathbf{x})^T \Upsilon D(\mathbf{x}) \right\} e(t). \end{aligned}$$

Depending on *a priori* information on behavior of the master system or the nonlinear term f , the above results can be restated. In Section II-B, we consider the case in which the error dynamics can be described by a polytopic system.

B. Polytopic Formulation

The conditions derived above contain a function matrix $D(\mathbf{x})$ in LMI constraints. Since $\mathbf{x}$ is a trajectory of the master system which shows chaotic behavior, $\mathbf{x}$ is bounded and in turn $D(\mathbf{x})$ is also a bounded matrix. Under the assumption that each element of $D(\mathbf{x})$ is bounded, the approximated error system (8) can be rewritten by the following polytopic system:

$$\dot{\mathbf{e}}(t) = A_0 \mathbf{e}(t) + \sum_{i=1}^m A_i \mathbf{e}(t - \tau_i) + \sum_{j=1}^k p_j(\mathbf{x}) D_j \mathbf{e}(t) \quad (14)$$

where D_j are constant matrices and $p_j(\mathbf{x}) \in [0, 1]$ are polytopic coordinates satisfying the convex sum property $\sum_{j=1}^k p_j(\mathbf{x}) = 1$.

Lemma 2.4: If there exists a symmetric and positive-definite matrix $P > 0$ and positive-definite matrices Γ_k , $0 \leq k \leq m$, of dimension $n \times n$ satisfying

$$\begin{aligned} & (\bar{A} + D_\ell)^T P + P(\bar{A} + D_\ell) \\ & + \sum_{i=1}^m \tau_i \left\{ (\bar{A} + D_\ell)^T \Gamma_0 (\bar{A} + D_\ell) \right. \\ & \left. + \sum_{j=1}^m A_j^T \Gamma_j A_j + P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} \right) A_i^T P \right\} < 0 \end{aligned} \quad (15)$$

for $\ell = 1, \dots, k$, where $\bar{A} = \sum_{i=0}^m A_i$, then the zero solution of (6) is asymptotically stable.

Proof: Since system (14) can be rewritten by

$$\dot{\mathbf{e}}(t) = \sum_{j=1}^k p_j(\mathbf{x}) \left((A_0 + D_j) \mathbf{e}(t) + \sum_{i=1}^m A_i \mathbf{e}(t - \tau_i) \right)$$

by using vertex systems, we can obtain the following *polytopic* linear differential inclusion (PLDI):

$$\dot{\mathbf{e}}(t) \in \text{Co} \left\{ (A_0 + D_1) \mathbf{e} + \sum_{i=1}^m A_i \mathbf{e}_{\tau_i}, \dots, (A_0 + D_k) \mathbf{e} + \sum_{i=1}^m A_i \mathbf{e}_{\tau_i} \right\} \quad (16)$$

where Co denotes a convex envelope. Therefore, if the vertex systems are asymptotically stable, any systems belonging to the PLDI are also asymptotically stable. Setting $Q_0 = (A_0 + D_j)^T \Gamma_0 (A_0 + D_j)$ and $R = 0$ in the proof of Lemma 2.1, a sufficient condition for each vertex system to be asymptotically stable is given by (15). $\square$

In the same way, we can obtain the counterpart of Theorem 2.2 for system (14).

Theorem 2.5: Let $\tau_i > 0$ for $i = 1, \dots, m$. If there exist a symmetric and positive-definite matrix $P > 0$ and a matrix Y such that the following LMI holds for each $\ell = 1, \dots, k$:

$$\begin{bmatrix} M_{11} + M_{11}^T & M_{12} & M_{13} \\ M_{12}^T & M_{22} & 0 \\ M_{13}^T & 0 & M_{33} \end{bmatrix} < 0 \quad (17)$$

where

$$\begin{aligned} M_{11} &= P(\bar{A} + D_\ell) + YC \\ M_{12} &= ((A_0 + D_\ell)^T P, \dots, A_m^T P + C^T Y^T) \\ M_{13} &= (PA_1, \dots, PA_m + YC) \\ M_{22} &= \frac{-1}{\sum_{i=1}^m \tau_i} \text{diag}(P, \dots, P) \\ M_{33} &= \frac{-1}{m+1} \text{diag} \left(\frac{1}{\tau_1} P \dots \frac{1}{\tau_m} P \right) \\ \bar{A} &= \sum_{i=0}^m A_i \end{aligned}$$

then the zero solution of system (6) is asymptotically stable and $K = P^{-1}Y$ is stabilizing gain matrix.

Proof: Replacing A_m with KC in (15) and setting $\Gamma_i = P$ and $Y = PK$, we obtain LMI (17). $\square$

C. Lur  Type Delay Systems

We consider the case where the prediction error $\mathbf{e}(t) = \mathbf{z}(t - \tau_m) - \mathbf{x}(t)$ is given by the following Lur  type difference-differential equation instead of (6):

$$\dot{\mathbf{e}}(t) = A_0 \mathbf{e}(t) + \sum_{i=1}^m A_i \mathbf{e}(t - \tau_i) + B \phi(C\mathbf{x}, C\mathbf{e}) \quad (18)$$

$$y(t) = C\mathbf{x}(t) \quad (19)$$

where $y \in \mathbb{R}$, $B \in \mathbb{R}^{n \times 1}$, $C \in \mathbb{R}^{1 \times n}$ and $\phi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$. In addition, instead of the continuity assumption on f , we assume that function $\phi(y, \hat{y})$ satisfies the following sector condition: There exists a $\alpha > 0$ such that $0 \leq \phi(y, \hat{y}) \leq \alpha \hat{y}$ for all $y \in \{y = C\mathbf{x} | \mathbf{x} \in \Omega\}$.

In this case, Lemma 2.1 is restated as follows.

Lemma 2.6: Let $\tau_i > 0$ for $i = 1, \dots, m$. If there exists a symmetric and positive-definite matrix $P > 0$, positive definite matrices Γ_i , $i = 0, \dots, m$ and Υ and a positive constant λ satisfying

$$\begin{bmatrix} \Theta & PB + \lambda \alpha C^T \\ B^T P + \lambda \alpha C & \sum_{i=1}^m \tau_i B^T \Upsilon B - 2\lambda \end{bmatrix} < 0 \quad (20)$$

where $\Theta = \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T \Gamma_j A_j + P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) A_i^T P \right)$, then the zero solution of the error dynamics is asymptotically stable.

Proof: This lemma is proved by replacing $D(\mathbf{x})\mathbf{e}(t)$ with $B\phi(C\mathbf{x}, C\mathbf{e})$ in the proof of Lemma 2.1

$$\begin{aligned} \frac{d}{dt}V(\mathbf{e}_t) &\leq \mathbf{e}(t)^T \{\bar{A}^T P + P\bar{A}\} \mathbf{e}(t) + 2\mathbf{e}(t)^T P B \phi(C\mathbf{x}, C\mathbf{e}) \\ &\quad + \sum_{i=1}^m \tau_i \phi(C\mathbf{x}, C\mathbf{e}) B^T \Upsilon B \phi(C\mathbf{x}, C\mathbf{e}) \\ &\quad + \mathbf{e}(t)^T \left\{ \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T \Gamma_j A_j \right. \right. \\ &\quad \left. \left. + P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) A_i^T P \right) \right\} \mathbf{e}(t). \end{aligned} \quad (21)$$

On the other hand, if the sector condition holds, the following inequality holds:

$$\phi(C\mathbf{x}, C\mathbf{e}) \{ \phi(C\mathbf{x}, C\mathbf{e}) - \alpha C\mathbf{e}(t) \} \leq 0. \quad (22)$$

Using (22), we obtain the following inequality:

$$\begin{aligned} \frac{d}{dt}V(\mathbf{e}_t) &\leq \mathbf{e}(t)^T \{\bar{A}^T P + P\bar{A}\} \mathbf{e}(t) + 2\mathbf{e}(t)^T P B \phi(C\mathbf{x}, C\mathbf{e}) \\ &\quad + \sum_{i=1}^m \tau_i \phi(C\mathbf{x}, C\mathbf{e}) B^T \Upsilon B \phi(C\mathbf{x}, C\mathbf{e}) \\ &\quad + \mathbf{e}(t)^T \left\{ \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T \Gamma_j A_j \right. \right. \\ &\quad \left. \left. + P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) A_i^T P \right) \right\} \mathbf{e}(t) \\ &\quad - 2\lambda \phi(C\mathbf{x}, C\mathbf{e}) \{ \phi(C\mathbf{x}, C\mathbf{e}) - \alpha C\mathbf{e}(t) \} \\ &= [\mathbf{e}(t)^T \quad \phi(C\mathbf{x}, C\mathbf{e})] \\ &\quad \times \begin{bmatrix} \Theta & P B + \lambda \alpha C^T \\ B^T P + \lambda \alpha C & \sum_{i=1}^m \tau_i B^T \Upsilon B - 2\lambda \end{bmatrix} \\ &\quad \times \begin{bmatrix} \mathbf{e}(t) \\ \phi(C\mathbf{x}, C\mathbf{e}) \end{bmatrix} \end{aligned} \quad (23)$$

where $\lambda \geq 0$ and $\Theta \triangleq \bar{A}^T P + P\bar{A} + \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T \Gamma_j A_j + P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) A_i^T P \right)$. Therefore, the matrix inequality $\begin{bmatrix} \Theta & P B + \lambda \alpha C^T \\ B^T P + \lambda \alpha C & \sum_{i=1}^m \tau_i B^T \Upsilon B - 2\lambda \end{bmatrix} < 0$ holds, then $V(\mathbf{e}_t) < 0$ and this means that the zero solution of system (18) is asymptotically stable. $\square$

Remark 2.7: If $\tau_i = 0$ for $i = 1, \dots, m$, the inequality (20) is reduced to

$$\begin{bmatrix} \bar{A}^T P + P\bar{A} & P B + \lambda \alpha C^T \\ B^T P + \lambda \alpha C & -2\lambda \end{bmatrix} < 0. \quad (24)$$

This corresponds to the stability criterion for Lur  systems by using a Lur -Postnikov type Lyapunov function.

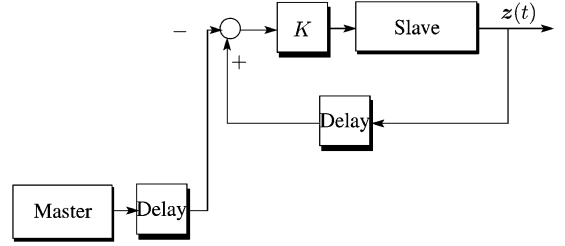


Fig. 2. Synchronization problem with communication delays.

Remark 2.8: Yal in *et al.* [12] have considered a synchronization problem of Lur  system with communication delay (Fig. 2) and have derived a similar LMI condition for the error system to be asymptotically stable by using a Lyapunov–Krasovskii functional. This means that if the master system does not contain time delay, the stability property of the prediction error in our prediction scheme is equivalent to the one of the synchronization error in the master–slave synchronization of systems coupled with communication delay. On the other hand, if the master system itself has time delays, the condition derived by Yal in *et al.* cannot be applied into such systems. In addition, we must note that in the synchronization problem of the coupled systems with communication delay, the output of the slave system is the estimated value $\hat{\mathbf{x}}(t)$ of the state $\mathbf{x}(t)$ of the master system, while, in our framework, the output $\mathbf{z}(t)$ of the slave system means an estimation of the predicted value $\mathbf{x}(t + \tau_m)$ of the state $\mathbf{x}(t)$ of the master system.

Using Lemma 2.6, we obtain the following asymptotic stabilizability condition for system (18).

Theorem 2.9: Let $\tau_i > 0$ for $i = 1, \dots, m$. If there exist a symmetric, positive-definite matrix $P > 0$ and a matrix Y such that the following holds:

$$\begin{bmatrix} M_{11} + M_{11}^T & M_{12} & M_{13} & M_{14} \\ M_{12}^T & M_{22} & 0 & 0 \\ M_{13}^T & 0 & M_{33} & 0 \\ M_{14}^T & 0 & 0 & M_{44} \end{bmatrix} < 0 \quad (25)$$

where

$$\begin{aligned} M_{11} &= P\bar{A} + YC \\ M_{12} &= (A_0^T P, \dots, A_m^T P + C^T Y^T) \\ M_{13} &= (P A_1, \dots, P A_m + YC) \\ M_{14} &= P B + \lambda \alpha C^T \\ M_{22} &= \frac{-1}{\sum_{i=1}^m \tau_i} \text{diag}(P, \dots, P) \\ M_{33} &= \frac{-1}{m+2} \text{diag}\left(\frac{1}{\tau_1} P, \dots, \frac{1}{\tau_m} P\right) \\ M_{44} &= \sum_{i=1}^m \tau_i B^T P B - 2\lambda \end{aligned}$$

then the zero solution of system is asymptotically stable and $K = P^{-1}Y$ is a stabilizing gain matrix.

Proof: Replacing A_m with $A_m + KC$ in Θ of inequality (20) yields

$$\begin{aligned} \Theta = & \bar{A}^T P + C^T K^T P + P \bar{A} + PKC \\ & + \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T \Gamma_j A_j \right. \\ & \quad \left. + (A_m + KC)^T \Gamma_m (A_m + KC) \right) \\ & + \sum_{i=1}^m \tau_i P A_i \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) A_i^T P \\ & + \tau_m P (A_m + KC) \left(\sum_{j=0}^m \Gamma_j^{-1} + \Upsilon^{-1} \right) \\ & \times (A_m + KC)^T P. \end{aligned}$$

By choosing as $\Gamma_i = \Upsilon = P$, Θ can be rewritten by

$$\begin{aligned} \Theta = & \bar{A}^T P + C^T K^T P + P \bar{A} + PKC \\ & + \sum_{i=1}^m \tau_i \left(\sum_{j=0}^m A_j^T P A_j (A_m + KC)^T P (A_m + KC) \right) \\ & + (m+2) \left(\sum_{i=1}^m \tau_i P A_i P^{-1} A_i^T P \right. \\ & \quad \left. + \tau_m P (A_m + KC) P^{-1} (A_m + KC)^T P \right). \end{aligned}$$

Setting $Y = PK$, we obtain LMI (25). $\square$

III. MAXIMUM PREDICTION TIME

As can be seen from the discussion above, we can obtain a gain K of the predictor by using Theorem 2.2, Theorem 2.5, or Theorem 2.9. The obtained results, however, are based on a delay-dependent stability and/or stabilizability by using a Lyapunov–Krasovskii functional. Therefore, we can expect that there exists an upper bound for the time delay τ_m that the prediction error can converge to 0. Voss [5] has shown a numerically estimated stability diagram with respect to the coupling gain and a time delay for the coupled Rössler system, but there is no theoretical discussion. In this section, we derive the available length of the prediction in the case of $m = 1$, that is the maximum time delay for synchronization of $z(t - \tau_1)$ and $\mathbf{x}(t)$ to be accomplished. For the purpose, we consider the prediction error dynamics (6) with $m = 1$. i.e.,

$$\dot{\mathbf{e}} = A_0 \mathbf{e} + A_1 \mathbf{e}_{\tau_1} + \phi(\mathbf{x}, \mathbf{e}). \quad (26)$$

The linearization system is also given by

$$\dot{\mathbf{e}} = A_0 \mathbf{e} + A_1 \mathbf{e}_{\tau_1} + D(\mathbf{x}) \mathbf{e}. \quad (27)$$

Then, we consider the available prediction time of the proposed prediction scheme and show an upper bound of the time delay by using the sufficient condition for asymptotic stability of the prediction error dynamics.

Rewriting Lemma 2.1, we obtain the following theorem.

Theorem 3.1: If there exist symmetric, positive-definite matrices $P > 0$, $Q_i > 0$, $i = 0, 1$ and R such that the following holds for all $\mathbf{x} \in \Omega$:

$$\begin{bmatrix} \Psi(\mathbf{x}) & P A_1 A_0 & P A_1 A_1 & P A_1 \\ A_0^T A_1^T P & -Q_0 & 0 & 0 \\ A_1^T A_1^T P & 0 & -Q_1 & 0 \\ A_1^T P & 0 & 0 & -R \end{bmatrix} < \begin{bmatrix} \Gamma & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (28)$$

$$\Gamma < \eta \{ -(\bar{A} + D(\mathbf{x}))^T P - P(\bar{A} + D(\mathbf{x})) \} \quad (29)$$

where $\Psi(\mathbf{x}) \triangleq Q_0 + Q_1 + D(\mathbf{x})^T R D(\mathbf{x})$ and $\eta = 1/\tau_1$, then the zero solution of (26) is asymptotically stable.

Proof: In the same way as in the proof of Lemma 2.1, we obtain the following inequality:

$$\begin{aligned} \dot{V} \leq & \mathbf{e}^T \left\{ (\bar{A} + D(\mathbf{x}))^T P + P(\bar{A} + D(\mathbf{x})) \right. \\ & + \tau Q_0 + \tau Q_1 + \tau_1 P (A_1 A_0) Q_0^{-1} (A_1 A_0)^T P \\ & + \tau_1 P (A_1 A_1) Q_1^{-1} (A_1 A_1)^T P \\ & + \tau_1 P A_1 R^{-1} A_1^T P \mathbf{e}(t) \\ & \left. + \tau_1 D(\mathbf{x})^T R D(\mathbf{x}) \right\} \mathbf{e}(t). \end{aligned}$$

Therefore, $\dot{V} < 0$ is equivalent to the following LMI:

$$\begin{bmatrix} \Xi + \tau_1 \Psi(\mathbf{x}) & \tau_1 P A_1 A_0 & \tau_1 P A_1 A_1 & \tau_1 P A_1 \\ \tau_1 A_0^T A_1^T P & -\tau_1 Q_0 & 0 & 0 \\ \tau_1 A_1^T A_1^T P & 0 & -\tau_1 Q_1 & 0 \\ \tau_1 A_1^T P & 0 & 0 & -\tau_1 R \end{bmatrix} < 0$$

where $\Xi = (\bar{A} + D(\mathbf{x}))^T P + P(\bar{A} + D(\mathbf{x}))$ and $\Psi(\mathbf{x}) = Q_0 + Q_1 + D(\mathbf{x})^T R D(\mathbf{x})$. This LMI is rewritten as

$$\begin{bmatrix} \Psi(\mathbf{x}) & P A_1 A_0 & P A_1 A_1 & P A_1 \\ A_0^T A_1^T P & -Q_0 & 0 & 0 \\ A_1^T A_1^T P & 0 & -Q_1 & 0 \\ A_1^T P & 0 & 0 & -R \end{bmatrix} < \frac{1}{\tau_1} \begin{bmatrix} -\Xi & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Furthermore, this LMI is equivalent to the following inequalities:

$$\begin{bmatrix} \Psi(\mathbf{x}) & P A_1 A_0 & P A_1 A_1 & P A_1 \\ A_0^T A_1^T P & -Q_0 & 0 & 0 \\ A_1^T A_1^T P & 0 & -Q_1 & 0 \\ A_1^T P & 0 & 0 & -R \end{bmatrix} < \begin{bmatrix} \Gamma & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Gamma < -\eta \{ (\bar{A} + D(\mathbf{x}))^T P + P(\bar{A} + D(\mathbf{x})) \}.$$

If $\dot{V} < 0$ holds, the zero solution of (27) is asymptotically stable. As a result, from Lyapunov's indirect method, we conclude that the zero solution of (26) is also asymptotically stable. $\square$

To get an upper bound of the time delay τ_1 for which the prediction error dynamics (26) remains asymptotically stable, one only needs to minimize η in LMI (28). Thus, the optimization problem that determines the upper bound of τ_1 can be stated as follows:

$$\min_{\eta>0, P>0, Q_0>0, Q_1>0, R>0, \Gamma>0} \eta$$

subject to LMI (28) and LMI (29). As a result, we obtain the following theorem.

Theorem 3.2: Let η_0 be the optimal solution of the following standard generalized eigenvalue problem (GEVP):

$$\min_{\eta>0, P>0, Q_0>0, Q_1>0, R>0, \Gamma>0} \eta$$

subject to LMI (28) and LMI (29) for all $\mathbf{x} \in \Omega$. Then for any $\tau_1 \in [0, 1/\eta_0]$, the prediction error dynamics (26) is asymptotically stable.

IV. EXAMPLES

This section shows numerical examples in order to verify the effectiveness of the proposed prediction scheme and the derived criteria.

A. Example 1: FitzHugh-Nagumo Neuron Model

This example focuses on the verification of Lemma 2.1 and Theorem 3.2. We consider the following FitzHugh–Nagumo neuron model [17]:

$$\begin{cases} \dot{x}_1(t) = -x_1(x_1 - a)(x_1 - 1) - x_2 + I(t) \\ \dot{x}_2(t) = \varepsilon(x_1 - bx_2) \\ y(t) = x_1(t) \end{cases} \quad (30)$$

where a , b , and ε are constants, and $I(t) \in [0.02, 0.04]$ is an external random forcing which is a piecewise constant. The parameters are $a = 0.139$, $b = 2.54$ and $\varepsilon = 0.008$. The predictive synchronization of this neuron model has been reported by Ciszak *et al.* [17], but they did not discuss on the stability region of the pair (k, τ) . In this section, we calculate the stability region by using results from the above sections. For system (30), we construct the following predictor:

$$\begin{cases} \dot{z}_1(t) = -z_1(z_1 - a)(z_1 - 1) - z_2 + I_0(t) \\ \quad + k(y(t) - z_1(t - \tau)) \\ \dot{z}_2(t) = \varepsilon(z_1 - bz_2) \end{cases}$$

where I_0 is the median of $I(t)$. The dynamics of the prediction error $\mathbf{e} = \mathbf{z}_\tau - \mathbf{x}$ is given by

$$\dot{\mathbf{e}} = \begin{bmatrix} -a & -1 \\ \varepsilon & -\varepsilon b \end{bmatrix} \mathbf{e} + \begin{bmatrix} \Delta(\mathbf{x}, \mathbf{e}) \\ 0 \end{bmatrix} + \begin{bmatrix} I_0(t - \tau) - I(t) \\ 0 \end{bmatrix} + \begin{bmatrix} -k & 0 \\ 0 & 0 \end{bmatrix} \mathbf{e}_\tau \quad (31)$$

where $\Delta(\mathbf{x}, \mathbf{e}) \triangleq -e_1^3 + (-3x_1 + a + 1)e_1^2 + \{2(a+1)x_1 - 3x_1^2\}e_1$ and $\mathbf{e}_\tau \triangleq \mathbf{e}(t - \tau)$. The dynamics linearized around $\mathbf{e} = 0$ is obtained by

$$\dot{\mathbf{e}} = \begin{bmatrix} -a + \bar{\Delta} & -1 \\ \varepsilon & -\varepsilon b \end{bmatrix} \mathbf{e} + \begin{bmatrix} I_0(t - \tau) - I(t) \\ 0 \end{bmatrix} + \begin{bmatrix} -k & 0 \\ 0 & 0 \end{bmatrix} \mathbf{e}_\tau$$

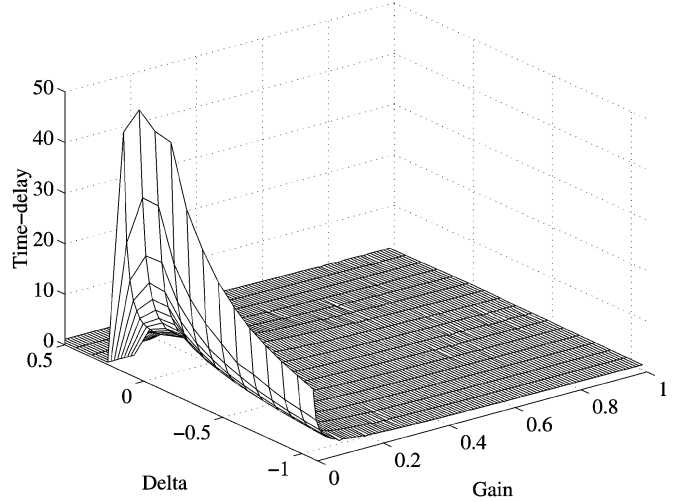


Fig. 3. Sufficient condition of stability of example 1.

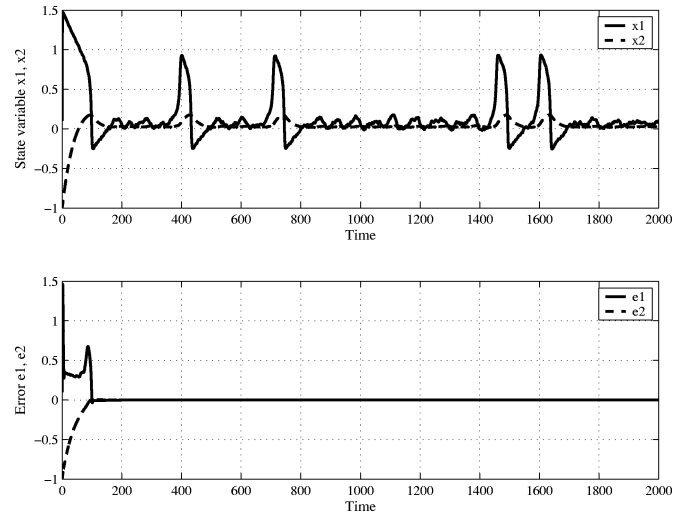


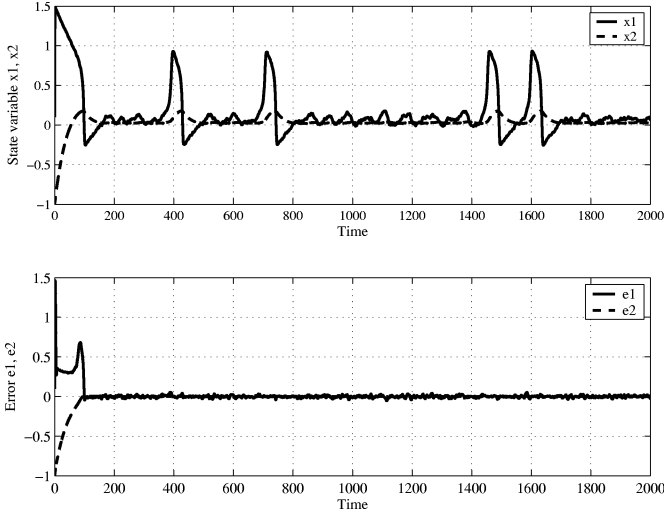
Fig. 4. Behaviors of $\mathbf{x}$ and $\mathbf{e}$ in the case of $I(t) = I_0(t - \tau)$ ($(k, \tau) = (0.25, 2)$).

where $\bar{\Delta} = 2(a+1)x_1 - 3x_1^2$. Using the boundness of x_1 and applying Theorem 3.2, we can calculate a stability region of (k, τ) for all $\mathbf{x} \in \Omega = \{(x_1, x_2) | -1.1 < \bar{\Delta} < 0.44\}$. Fig. 3 shows the bound of the stability region of (k, τ) with respect to each $\bar{\Delta}$. The derived region shows a sufficient condition with respect to $(k, \bar{\Delta}, \tau)$ under which the prediction error dynamics is asymptotically stable. If $I_0(t)$ equals the future value of the external force $I(t)$ of the master system, i.e., $I_0(t) = I(t + \tau)$, (31) is rewritten as

$$\dot{\mathbf{e}} = \begin{bmatrix} -a + \bar{\Delta} & -1 \\ \varepsilon & -\varepsilon b \end{bmatrix} \mathbf{e} + \begin{bmatrix} -k & 0 \\ 0 & 0 \end{bmatrix} \mathbf{e}_\tau.$$

Fig. 4 shows the behaviors of $\mathbf{x}$ and the prediction error $\mathbf{e}(t)$ in the case of $I_0(t - \tau) = I(t)$. We know that the prediction error exactly converges to zero.

Next, we show a simulation results in the case of $I_0 = 0.03$, which is the median of $I(t)$, in Fig. 5. Although the predicted value is affected by the difference of the external forces, the behavior can be predicted sufficiently.


 Fig. 5. Behaviors of $\mathbf{x}$ and $\mathbf{e}$ in the case of $I_0 = 0.03$.

B. Example 2: Time-Delay Chua's Oscillator

Next we show an example in the case where the prediction error dynamics is described by the Lur  type. We consider the following Chua's circuit with time delay [16].

$$\begin{cases} \dot{x}_1(t) = \alpha(x_2 - x_1 - h(x_1)) \\ \dot{x}_2(t) = x_1 - x_2 + x_3 \\ \dot{x}_3(t) = -\beta x_2 - \gamma x_3 - \beta \varepsilon \sin(\eta x_1(t - \tau_1)) \\ y(t) = x_1(t) \end{cases} \quad (32)$$

where the nonlinear function $h(x_1)$ is given by

$$h(x_1) = bx_1 + \frac{1}{2}(a - b)(|x_1 + 1| - |x_1 - 1|)$$

with $\alpha = 10$, $\beta = 19.53$, $\gamma = 0.1636$, $a = -1.4325$, $b = -0.7831$, $\eta = 0.5$, $\varepsilon = 0.2$ and $\tau_1 = 0.019$. The chaotic behavior of this system is shown in Fig. 6. Cruz-Hern ndez has studied synchronization of this system and applied the method for secure communication. In this paper, we attempt to construct the predictor proposed above for this system. This system can be rewritten by

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \begin{bmatrix} -\alpha(1+\beta) & \alpha & 0 \\ 1 & -1 & 1 \\ 0 & -\beta & -\gamma \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} -\alpha(a-b) \\ 0 \\ 0 \end{bmatrix} \\ &\quad \times \frac{1}{2}(|x_1 + 1| - |x_1 - 1|) + \begin{bmatrix} 0 \\ 0 \\ -\beta \end{bmatrix} \varepsilon \sin(\eta x_1(t - \tau_1)) \\ &\triangleq A_0 \mathbf{x}(t) + B\varphi(x_1) + B_1 \xi(x_1(t - \tau_1)) \\ y(t) &= [1 \ 0 \ 0] \mathbf{x}(t) \\ &\triangleq C\mathbf{x}(t) \end{aligned}$$

where $\varphi(x_1) = (1/2)(|x_1 + 1| - |x_1 - 1|)$ and $\xi(x_1(t)) = \varepsilon \sin(\eta x_1(t))$. Although this system cannot be described by (4) because one of nonlinear terms is a function of the delayed state $x_1(t - \tau_1)$, by combining the prediction scheme of (3), a predictor whose zero solution is asymptotically stable can be designed through our proposed design procedure.

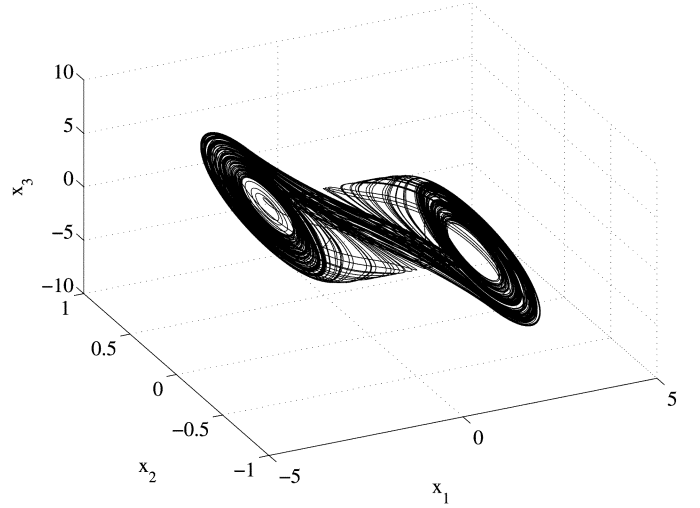
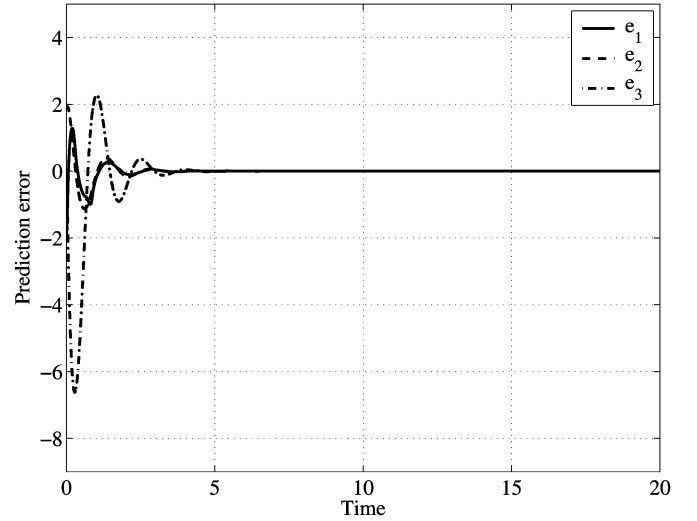


Fig. 6. Phase portrait of time-delay Chua's circuit (32).


 Fig. 7. Prediction error $\mathbf{e}(t) = \mathbf{z}(t - \tau_1) - \mathbf{x}(t)$.

If we construct the following predictor:

$$\Sigma_s : \begin{cases} \dot{\mathbf{z}}(t) = A_0 \mathbf{z}(t) + B\varphi(z_1) + B_1 \xi(y(t)) \\ \quad + K\{C\mathbf{z}(t - \tau_1) - y(t)\} \\ \mathbf{z}(t) = \mathbf{z}_0, \quad t \in [-2\tau_1, 0] \end{cases}$$

then the dynamics of the prediction error $\mathbf{e}(t) = \mathbf{z}(t - \tau_1) - \mathbf{x}(t)$ is given by

$$\begin{aligned} \dot{\mathbf{e}}(t) &= A_0 \mathbf{e}(t) + K C \mathbf{e}(t - \tau_1) \\ &\quad + B\{\varphi(e_1(t) + x_1(t)) - \varphi(x_1(t))\} \\ &\triangleq A_0 \mathbf{e}(t) + K C \mathbf{e}_{\tau_1} + B\phi(x_1(t), e_1(t)) \end{aligned} \quad (33)$$

where $\mathbf{e}_{\tau_1} \triangleq \mathbf{e}(t - \tau_1)$. Please note that this system can be recognized as a Lur  type system because there exists a real number $\alpha_0 = 1$ satisfying the following sector condition:

$$0 \leq \phi(x_1, e_1) \leq \alpha_0 e_1.$$

Therefore, the assumption of Section II-C is fulfilled. According to Theorem 2.9, we obtain a stabilizing gain $K = P^{-1}Y =$

$[-12.1986, -2.4324, 3.5627]^T$ by solving P and Y satisfying the LMI (25). Considering $\mathbf{x}(0) = [1, -2, 0]^T$ and $\mathbf{x}(\theta) = 0$ for $\theta < 0$ as the initial condition of the master system and $\mathbf{z}(\theta) = 0$ for $\theta < 0$ as that of the slave system, the behavior of the prediction error $\mathbf{e}(t) = \mathbf{z}(t - \tau_1) - \mathbf{x}(t)$ is depicted in Fig. 7. Since the prediction error converges to the origin, we know that the slave system works effectively as a predictor of the master system. As a result, by applying this predictor into a chaotic secure communication, we can reduce a decoding time because the anticipation time can be used to decode the received message in real time.

V. CONCLUSION

In this paper, we have considered state prediction for possibly chaotic or complex systems with time delay using a synchronization of the coupled chaotic systems. Especially we have focused on a predictor design using a static linear coupling as the first step of the theoretical research on this field. As a result, we derived a sufficient condition for the prediction error to converge to zero by using a Lyapunov–Krasovskii functional. In addition, we derived an upper bound of available time delay for the proposed prediction method by using a standard generalized eigenvalue problem. Since the obtained result here is a sufficient condition, it may be too restrictive for anticipating synchronization. Moreover, throughout this paper, we assume that the model of the chaotic/complex system is exactly known and an identical copy of the system can be constructed. In a practical situation, however, chaotic systems to be coupled may have some mismatch and uncertainties of the parameters. Therefore, it will be required to derive a condition for anticipating synchronization of systems with model uncertainties and perturbation. This will be the subject of future research.

REFERENCES

- [1] E. Ott, *Chaos in Dynamical Systems*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2002.
- [2] M. J. Ogorzalek, "Taming chaos—Part I: Synchronization," *IEEE Trans. Circuits Syst. I, Fundam. Theory Appl.*, vol. 40, no. 10, pp. 693–699, Oct. 1993.
- [3] K. Pyragas, "Synchronization of coupled time-delay systems: Analytical estimations," *Phys. Rev. E*, vol. 58, no. 3, pp. 3067–3071, 1998.
- [4] H. Nijmeijer, "A dynamical control view on synchronization," *Phys. D*, vol. 154, pp. 219–228, 2001.
- [5] H. U. Voss, "Anticipating chaotic synchronization," *Phys. Rev. E*, vol. 61, no. 5, pp. 5115–5119, 2000.
- [6] —, "Real-time anticipation of chaotic states of an electronic circuit," *Int. J. Bifurc. Chaos*, vol. 12, no. 7, pp. 1619–1625, 2002.
- [7] C. Masoller, "Anticipation in the synchronization of chaotic semiconductor laser with optical feedback," *Phys. Rev. Lett.*, vol. 86, no. 13, pp. 2782–2785, 2001.
- [8] S. Sivaprakasam, E. M. Shahverdiev, P. S. Spencer, and K. A. Shore, "Experimental demonstration of anticipating synchronization in chaotic semiconductor lasers with optical feedback," *Phys. Rev. Lett.*, vol. 87, no. 15, pp. 154 101–154 101, 2001.
- [9] J. K. Hale and S. M. V. Lunel, *Introduction to Functional Differential Equations*. New York: Springer-Verlag, 1993.
- [10] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan, *Linear Matrix Inequalities in Systems and Control Theory*. Philadelphia, PA: SIAM, 1994.
- [11] R. E. Skelton, T. Iwasaki, and K. Grigoriadis, *A Unified Algebraic Approach to Linear Control Design*. New York: Taylor & Francis, 1998.
- [12] M. E. Yalçın, J. A. K. Suykens, and J. Vandewalle, "Master-slave synchronization of Luré systems with time delay," *Int. J. Bifurc. Chaos*, vol. 11, no. 6, pp. 1707–1722, 2001.
- [13] M. S. Mahmoud, *Robust Control and Filtering for Time-Delay Systems*. New York: Dekker, 2000.
- [14] V. Kolmanovskii and A. Myshkis, *Introduction to the Theory and Application of Functional Differential Equations*. Dordrecht, The Netherlands: Kluwer, 1999.
- [15] E.-K. Boukas and Z.-K. Liu, *Deterministic and Stochastic Time Delay Systems*. Berlin, Germany: Birkhäuser, 2001.
- [16] C. Cruz-Hernández, "Synchronization of time-delay Chua's oscillator with application to secure communication," *Nonlin. Dyn. Syst. Theory*, vol. 4, no. 1, pp. 1–13, 2004.
- [17] M. Ciszak, O. Calvo, C. Masoller, C. R. Mirasso, and R. Toral, "Anticipating the response of excitable systems driven by random forcing," *Phys. Rev. Lett.*, vol. 90, no. 20, pp. 204 102–204 102, 2003.



Toshiki Oguchi (M'96) received the B.S., M.S. and Dr. Eng. degrees in mechanical engineering from Tokyo Metropolitan University, Tokyo, Japan, in 1990, 1992, and 1995, respectively.

From 1995 to 1997 he was a Research Associate in the Department of Computer Science, National Defense Academy, Kanagawa, Japan. Since 1997, he has been with the Department of Mechanical Engineering, Graduate School of Engineering, Tokyo Metropolitan University, where he is currently an Associate Professor. In 2003, he was a Visiting Scientist in the Department of Mechanical Engineering, Eindhoven University of Technology, Eindhoven, The Netherlands. His current research interests include control theory for nonlinear systems with time delays, control of complex systems, and nonlinear systems theory.



Henk Nijmeijer (M'83–SM'91–F'00) received the M.Sc. and Ph.D. degrees in mathematics from the University of Groningen, Groningen, The Netherlands, in 1979 and 1983, respectively.

From 1983 until 2000, he was affiliated with the Department of Applied Mathematics, University of Twente, Enschede, The Netherlands. Since 1997 he was also part-time affiliated with the Department of Mechanical Engineering of the Eindhoven University of Technology, Eindhoven, The Netherlands. Since 2000, he is full-time faculty at the Eindhoven University of Technology, and chairs the Dynamics and Control section. He has published a large number of journal and conference papers, and several books, including *Nonlinear Dynamical Control Systems* (New York: Springer Verlag, 1990) with coauthor A. J. van der Schaft. He is Editor-in-Chief of the *Journal of Applied Mathematics*, Corresponding Editor of the *SIAM Journal on Control and Optimization*, and Board Member of the *International Journal of Control, Automatica*, *European Journal of Control*, *Journal of Dynamical Control Systems*, *SACTA*, *International Journal of Robust and Nonlinear Control*, and the *Journal of Applied Mathematics and Computer Science*.

Dr. Nijmeijer was awarded the IEE Heaviside premium in 1987.